

DESIGN SUMMARY

Running head: Table Logic Real-Time Tracker

Table Logic: Wireframe Design Summary

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for

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Note: This document is to be committed to the repository at the path:
/docs/wireframes/Design_Summary.pdf

1. Screen Design Decisions

Authentication Screen (Staff Login): The login interface was designed with strict minimalism to serve as a fast entry point for employees. To prevent scope creep regarding complex JWT backend routing in the MVP, the wireframe focuses solely on Employee ID and Password inputs, utilizing a simple frontend state toggle for authentication. Priority was placed on permanent, visible labels rather than vanishing placeholder text.

Customer Map (Dashboard): This screen acts as the read-only digital floor plan for patrons. We utilized a clear grid layout with high-contrast color coding (Green for Available, Red for Occupied). A specific design decision was made to include a centralized "Loading" state spinner, satisfying the HCD heuristic for Visibility of System Status while the frontend fetches live API data.

Host Control Dashboard: This is the core interactive hub where the primary application value is delivered. The layout mirrors the Customer Map but introduces interactive hover states. We explicitly removed "Reservations" from this screen's scope, replacing it with a "Current Waitlist" panel to align with our approved MVP and avoid complex time-based backend logic.

Table Details Modal: Rather than navigating to an entirely new page, clicking a table triggers a centered modal overlay. This displays the table ID, seat capacity, and the primary status toggle button. Crucially, an 'X' icon was placed in the top right corner to satisfy Nielsen's heuristic for User Control and Freedom, ensuring the host can escape the screen without accidentally changing the database state.

Filter & Summary Panel: Designed as a supporting secondary workflow, this side panel allows the host to quickly toggle views (e.g., "Show 4-Tops"). The design decision was made to handle this filtering entirely on the frontend via JavaScript array filtering, keeping the UI instantly responsive and reducing unnecessary database querying.

2. Accessibility (WCAG 2.1 AA) Implementations

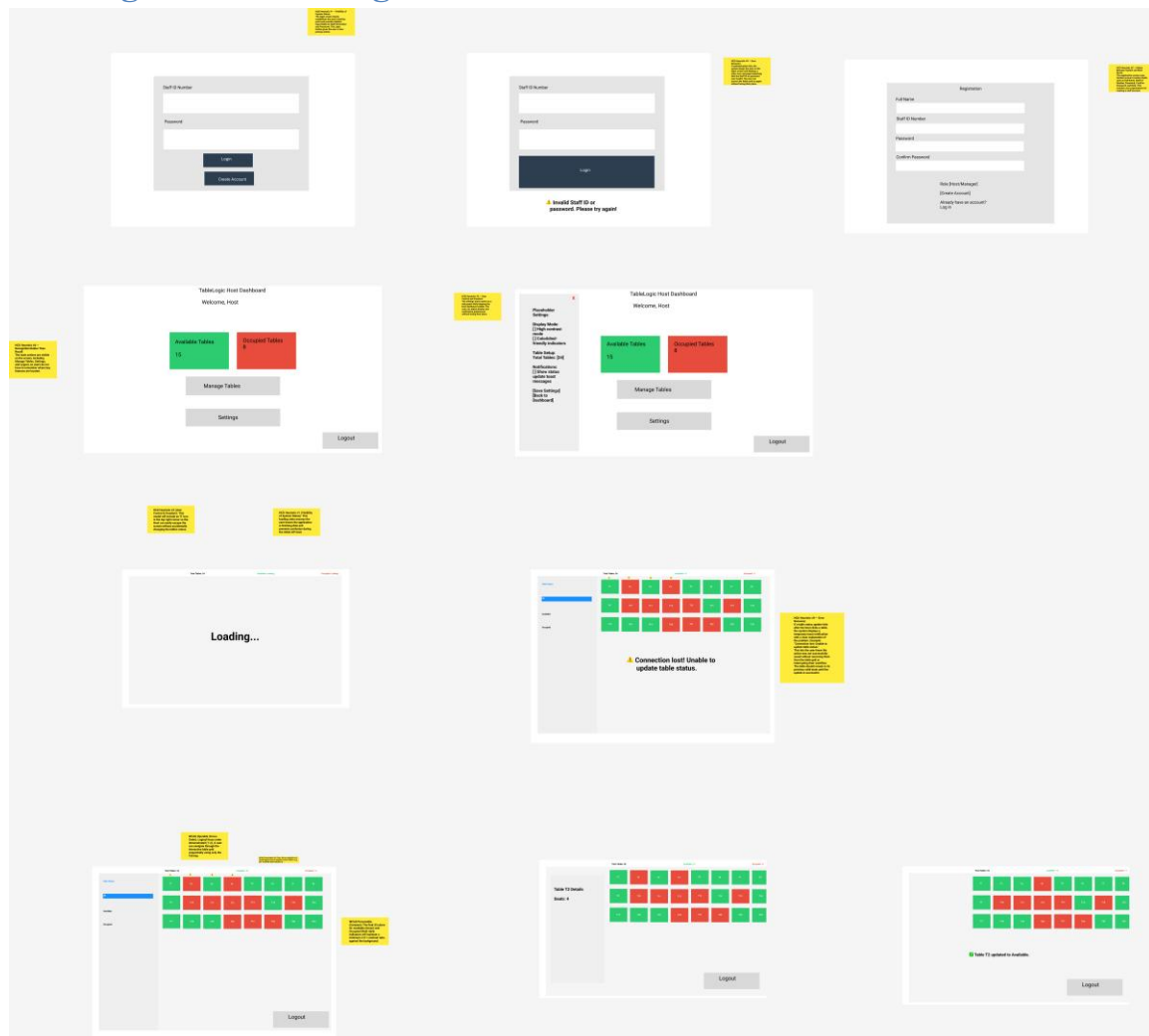
The wireframes were explicitly designed to meet the following WCAG standards (documented via Master QA sticky notes in the Figma file):

- **Color Contrast (Perceivable):** The final UI colors for the Green (Available) and Red (Occupied) table indicators were checked to ensure they maintain a minimum 4.5:1 contrast ratio against the background.

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- Alt Text (Perceivable): Descriptive alternative text annotations (e.g., [alt: "Available table indicator"]) were applied to all visual legend icons to support screen readers.
- Visible Form Labels (Understandable): The Staff Login screen completely avoids placeholder-only labels. "Staff ID Number" and "Password" exist as permanent text elements above their respective input boxes.
- Focus Order (Operable): The interactive table grid was designed with a logical, sequential focus order (annotated 1-4 on the wireframe) to ensure a user can navigate seamlessly using only the Tab key.
- Error States (Understandable): A Toast notification variant was designed for the Host Dashboard to provide clear, descriptive error recovery (e.g., "Connection lost: Unable to update table status") in the event of a failed database PATCH request.

3. Navigation Flow Diagram



The navigation flow is strictly linear from the Staff Login directly to the Host Dashboard. From the Host Dashboard, the user can branch outward to the Table Details Modal or the

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Filter Panel, but all paths safely return to the central dashboard to prevent the user from getting lost in deep menus.

4. Anticipated Changes After Usability Feedback (Week 9)

As we move toward high-fidelity prototypes and conduct user testing in Week 9, we expect to make several adjustments based on real-world feedback. First, we anticipate needing to modify the Red/Green color palette—perhaps by introducing distinct visual patterns (like crosshatching for occupied tables)—to better accommodate users with red-green color blindness. Second, we expect feedback regarding the "hit area" sizing on the Host Dashboard; if hosts are using the application on iPads during a busy dinner rush, the table squares may need to be enlarged to prevent "fat-finger" selection errors. Finally, the placement of the "End of Day Reset" button on the Filter panel may need to be moved behind an additional confirmation wall if usability testing reveals hosts are accidentally clicking it.